Questions numbered 2–10 correspond to Questions 1–9 in the assignment.

2. For $\mathbf{E} = \hat{x}\cos x - \hat{y}\sin x$,

   a) (6 pts)

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos x & -\sin x & 0 \end{vmatrix} = -\hat{z}\cos x,$$

$$\nabla \times (\nabla \times \mathbf{E}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & -\cos x \end{vmatrix} = -\hat{y}\sin x.$$

   b) (4 pts)Referring to the differential form of Gauss' law, we can write

$$\rho = \nabla \cdot \mathbf{D} = \epsilon_0 (\nabla \cdot \mathbf{E}),$$

   from which we obtain

$$\rho = \epsilon_0 \left[\left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} \right) \cdot (\hat{x}\cos x - \hat{y}\sin x) \right] = -\epsilon_0 \sin x \, \frac{\mathrm{C}}{\mathrm{m}^3}.$$

3. For $V(x, y, z) = 5z - 2x^2$ V in a certain region of space,

   a) (3 pts)$\mathbf{E} = -\nabla V = -\left(\widehat{x}\frac{\partial}{\partial x} + \widehat{y}\frac{\partial}{\partial y} + \widehat{z}\frac{\partial}{\partial z} \right)(5z - 2x^2) = 4x\,\widehat{x} - 5\widehat{z}\,\frac{\mathrm{V}}{\mathrm{m}}$,

   b) (3 pts)$\nabla \times \mathbf{E} = \begin{vmatrix} \widehat{x} & \widehat{y} & \widehat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4x & 0 & -5 \end{vmatrix} = 0,$

c) (2 pts)$\nabla \cdot \mathbf{E} = \left(\widehat{x}\frac{\partial}{\partial x} + \widehat{y}\frac{\partial}{\partial y} + \widehat{z}\frac{\partial}{\partial z}\right) \cdot (4x\,\widehat{x} - 5\hat{z}) = 4\ \frac{\mathrm{V}}{\mathrm{m}^2}$, and

d) (2 pts)$\rho = \nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon_0 \mathbf{E}) = \epsilon_0\,(\nabla \cdot \mathbf{E}) = 4\epsilon_0\ \frac{\mathrm{C}}{\mathrm{m}^3}$.

4. (10 pts)Starting with the left-handside of $\nabla \times (\nabla \times \mathbf{E}) = \nabla\,(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, we write

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= \nabla \times \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos x & -\sin x & 0 \end{vmatrix} \\ &= \nabla \times (0, 0, -\cos x) = (0, -\sin x, 0). \end{aligned}$$

Solving the right-handside of the equation, we obtain

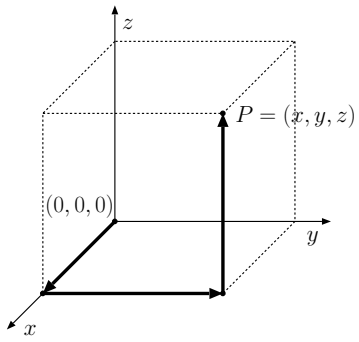
$$\begin{aligned} \nabla\,(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} &= \nabla\,(\nabla \cdot (\hat{x}\cos x - \hat{y}\sin x)) - \nabla^2\,(\hat{x}\cos x - \hat{y}\sin x) \\ &= \nabla\,(-\sin x) - (-\hat{\mathbf{x}}\cos \mathbf{x} + \hat{\mathbf{y}}\sin \mathbf{x}) \\ &= -\hat{x}\cos x + \hat{x}\cos x - \hat{y}\sin x \\ &= (0, -\sin x, 0). \end{aligned}$$

Consequently, we can see that $\nabla \times (\nabla \times \mathbf{E}) = \nabla\,(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ is verified since both sides have the same solutions.

5. (10 pts)In electrostatics, we generate a curl-free vector field $\mathbf{E}(x, y, z)$ if we take the gradient of a scalar function $V(x, y, z)$. Therefore, $\nabla \times \mathbf{E} = \nabla \times (-\nabla V) = 0$.

6. (10 pts)Maxwell equations for electrostatics state that $\nabla \times \mathbf{E} = 0$. Since $\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4 & 8z^2x & 0 \end{vmatrix} = -16zx\hat{x} + 8z^2\hat{z}$, $\mathbf{E}$ is not a possible electrostatic field.

7. (10 pts)Since the given field satisfies$\nabla \times \mathbf{E} = 0$, it is said to be an electrostatic field which can be indicated as $\mathbf{E} = -\nabla V$. Thus, the electrostatic potential V at any point $P = (x, y, z)$ can be calculated by total differential method, or performing a vector line integral using the path shown in the below figure.



Approach 1: For the total differential method, we start from the relation

$$\mathbf{E} = -\nabla V = -(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}). \tag{13}$$

From which we can get

$$\begin{aligned} -dV = \mathbf{E}\cdot d\mathbf{l} &= -(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}) \cdot (\hat{x}dx + \hat{y}dy + \hat{z}dz) \\ &= -(\frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz) = d(2x + 2y^2 - z^3). \end{aligned} \tag{14}$$

Therefore we can conclude that

$$V(x, y, z) = -(2x + 2y^2 - z^3) + C \tag{15}$$

for some constant C. Plugging in $V(0,0,0) = 1\,\text{V}$, we get $C = 1$, so we have:

$$V(x,y,z) = -(2x + 2y^2 - z^3) + 1. \tag{16}$$

Thus

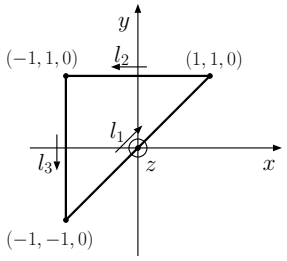
$$V(P) = V(1,2,3) = 18\,\text{V}.$$

Approach 2: Using the line integral wiht the path shown in the figure, we have:

$$\begin{aligned} V(P) &= -\int_0^P \mathbf{E}\cdot d\mathbf{l} + V(0) \\ &= -\int_0^x E_x(x,0,0)dx - \int_0^y E_y(x,y,0)dy - \int_0^z E_z(x,y,z)dz + V(0) \\ &= -(2x + 2y^2 - z^3) + 1. \end{aligned}$$

Therefore, the electrostatic potential at $(1,2,3)$ will be $V(1,2,3) = 18\,\text{V}$.

8. (10 pts)The triangular path defined in the problem is sketched in the below figure.



Referring to the hint given in the problem, we can write

$$\oint_C \mathbf{E}\cdot d\mathbf{l} = \int_{l_1} \mathbf{E}(x,x,0)\cdot d\mathbf{l}_1 + \int_{l_2} \mathbf{E}(1,\pm x,0)\cdot d\mathbf{l}_2 + \int_{l_3} \mathbf{E}(y,\mp 1,0)\cdot d\mathbf{l}_3.$$

Evaluating this equation for the field $\mathbf{E}$ with the $+$ sign, we obtain

$$\oint_C \mathbf{E}\cdot d\mathbf{l} = \int_{-1}^{1} (x\hat{x} + x\hat{y})\cdot(\hat{x}+\hat{y})dx + \int_{1}^{-1} (\hat{x} + x\hat{y})\cdot(\hat{x})dx + \int_{1}^{-1} (y\hat{x} - \hat{y})\cdot(\hat{y})dy = 0\,\text{V}.$$

Likewise, we calculate

$$\oint_C \mathbf{E}\cdot d\mathbf{l} = \int_{-1}^{1} (x\hat{x} - x\hat{y})\cdot(\hat{x}+\hat{y})dx + \int_{1}^{-1} (\hat{x} - x\hat{y})\cdot(\hat{x})dx + \int_{1}^{-1} (y\hat{x} + \hat{y})\cdot(\hat{y})dy = -4\,\text{V}$$

for the field $\mathbf{E}$ with the $-$ sign.

9. (10 pts)The voltage drop between the plates will be calculated as

$$V_p = V_{z=4} - V_{z=0} = -\int_0^4 \mathbf{E}\cdot d\mathbf{l} = -\int_0^4 \mathbf{E}\cdot\hat{z}dz = -\int_0^4 3\,dz = -12\,\text{V}.$$

and similarily for V_q

$$V_q = V_{z=0} - V_{z=4} = -V_p = 12\,\text{V}.$$

10.

a) (3 pts)One of the boundary conditions state that at any surface S, the normal component of the displacement field $\mathbf{D}$ can change by an amount equal to the surface charge density. Applying this boundary condition to the interface at $z = 5\,\text{m}$, the surface charge density is calculated as

$$\begin{aligned} \rho_s &= \hat{z} \cdot (\mathbf{D}\,|_{z=5^+} - \mathbf{D}\,|_{z=5^-}) \\ &= 4 - 6 = -2\,\frac{\text{C}}{\text{m}^2}. \end{aligned}$$

b) (3 pts)Another boundary condition also states that the tangential component of electric field $\mathbf{E}$ needs to be continuous along the surface S. Therefore, we can write

$$E_x\,|_{z=5^+} = E_x\,|_{z=5^-}\ ,$$

Also, we know that $D_z = 4\,\text{C/m}^2$ for $z > 5\,\text{m}$. Given that the space is vacuum, we can verify that $D_x|_{z=5^+} = D_x|_{z=5^-} = 3\,\frac{\text{C}}{\text{m}^2}$. Consequently, the displacement field $\mathbf{D}$ for the region $z > 5\,\text{m}$ can be expressed as

$$\mathbf{D} = 3\hat{x} + 4\hat{z}\,\frac{\text{C}}{\text{m}^2}.$$

c) (4 pts)Looking at the charge distribution, one can see that there is a charge density of $8\,\text{C/m}^2$ along the interface at $z = 0$. Thus, applying boundary conditions, we can write

$$\rho_s = \hat{z} \cdot (\mathbf{D}\,|_{z=0^+} - \mathbf{D}\,|_{z=0^-}),$$

from which we obtain $D_z|_{z=0^-} = -2\,\frac{\text{C}}{\text{m}^2}$. Given that the space is vacuum, we can also verify that $D_x|_{z=0^+} = D_x|_{z=0^-} = 3\,\frac{\text{C}}{\text{m}^2}$. Consequently, the displacement field $\mathbf{D}$ for the region $z < 5\,\text{m}$ can be expressed as

$$\mathbf{D} = 3\hat{x} - 2\hat{z}\,\frac{\text{C}}{\text{m}^2}.$$